

# Office Attendance System

Administrator Operations Manual

|               |                                             |
|---------------|---------------------------------------------|
| Document type | Administrator Manual                        |
| System        | ESP32 Fingerprint Attendance — Multi-Branch |
| Audience      | Branch administrators                       |
| Version       | 1.0                                         |

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**CHAPTER 1**

# Getting Started

What you need to know before you begin

## 1.1 What to Expect When the Device Arrives

When you receive your attendance device, it will be fully assembled and pre-configured with your branch's details. The only things you need to do before it is ready for daily use are: connect it to your office WiFi network, and enroll fingerprints for your employees. Everything else has been set up for you.

**Note**

The device has no battery. It must be plugged into a power source whenever it is in use. Keep it plugged in at all times during the work day.

## 1.2 What Has Already Been Done For You

The following have been completed before the device was sent to you:

- The backend system has been deployed and is fully operational.
- Your branch has been registered in the central system.
- A Google Sheet for your branch has been created and linked to the system.
- The device firmware has been flashed and is ready to use.
- The device has been configured with your branch ID.

## 1.3 Your Responsibilities as Administrator

As the branch administrator, your ongoing responsibilities are:

- **First-time setup:** Connect the device to WiFi and enroll the master finger.
- **Employee management:** Enroll new employees, remove employees who leave, and replace fingerprints when needed.
- **WiFi maintenance:** Reconfigure the WiFi connection if the office network changes.
- **Record keeping:** Monitor and correct attendance records in Google Sheets as needed.
- **Monthly reports:** Review the automated monthly summary email sent to your inbox.

## CHAPTER 2

## First-Time WiFi Setup

Connecting the device to your office network

### 2.1 Powering On the Device

Before anything else, the device needs power. Plug it into a wall socket or USB power source using the cable provided. Once powered on, the fingerprint sensor LED will begin glowing — this confirms the device is running.

On first boot, the device has no WiFi credentials saved and will enter setup mode automatically. The sensor LED will pulse **purple** to indicate it is waiting to be configured.

### 2.2 Connecting the Device to Office WiFi

- 1 On your phone or laptop, open your WiFi settings and look for a network named **Attendance-Setup**.
- 2 Connect to **Attendance-Setup**. The password is **setup1234**.
- 3 Once connected, open a web browser and go to: **http://192.168.4.1**. A setup page will appear.
- 4 Enter your office WiFi name (SSID) and password into the form.
- 5 Click **Save & Connect**. The device will save the credentials and restart automatically.
- 6 Reconnect your phone or laptop to your normal office WiFi after this step.



Screenshot: Attendance device WiFi setup page in browser

**■ Note**

If the setup page does not appear automatically after connecting to Attendance-Setup, type `http://192.168.4.1` manually into your browser's address bar.

**✗ Warning**

Do not share the Attendance-Setup password (setup1234) with employees. Anyone who knows it and has physical access to the device could reconfigure it.

## 2.3 Verifying the Connection

After the device restarts, check the sensor LED:

| LED state     | Meaning                                                                |
|---------------|------------------------------------------------------------------------|
| Steady blue   | Connected to WiFi successfully. Ready to use.                          |
| Steady purple | WiFi connection failed. Check your password and try again from Step 1. |

**✓ Tip**

If the LED stays purple after restarting, the most common cause is a wrong WiFi password. Repeat Section 2.2 and re-enter the credentials carefully.

## CHAPTER 3

## Enrolling the Master Finger

Setting up the administrator fingerprint

### 3.1 What the Master Finger Does

The master finger is a special administrator fingerprint enrolled on the device. Its only purpose is to trigger the WiFi setup mode — useful when you need to change the office WiFi network in the future. When the device recognises the master finger, it flashes yellow five times and launches the WiFi setup portal.

**Note**

The master finger does **not** record attendance. Scanning it will never create an entry in the attendance sheet. It is purely an admin tool.

The master finger should be enrolled using your own fingerprint, or the fingerprint of whoever is responsible for maintaining the device.

### 3.2 How to Enroll the Master Finger

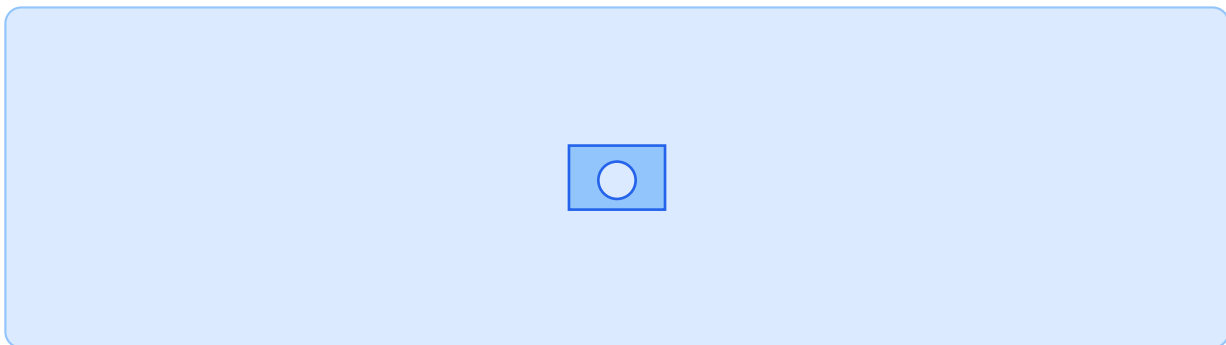
Enrollment is managed from the Google Sheet that was set up for your branch. You will need edit access to that sheet. Open it in Google Sheets before proceeding, and make sure the device is powered on and its LED is **steady blue**.

- 1 Open your branch spreadsheet in Google Sheets.
- 2 Click on the tab labelled **Enrollment** at the bottom of the sheet.
- 3 Click on the first empty row below the header row.
- 4 Fill in the columns as follows:

| Field    | Description                                                        |
|----------|--------------------------------------------------------------------|
| FingerID | Leave this blank. The device will fill it in after enrollment.     |
| UniqueID | Enter any short identifier for the master, e.g. <b>MASTER-01</b> . |
| Name     | Enter a label, e.g. <b>Admin Master</b> .                          |
| Role     | Type exactly: <b>master</b>                                        |
| Command  | Type exactly: <b>register</b>                                      |

| Field     | Description                                            |
|-----------|--------------------------------------------------------|
| Status    | Leave this blank.                                      |
| Note      | Leave this blank.                                      |
| CreatedAt | Optional. You may enter today's date for your records. |

- 5 Within about 10 seconds, the device LED will turn **solid yellow**. This means it has picked up the enrollment request and is waiting for a finger.
- 6 Place your finger firmly on the sensor and hold it still until the LED changes.
- 7 Lift your finger when prompted (the LED will go off briefly), then place **the same finger** on the sensor again for the second scan.
- 8 If successful, the LED will flash **green**. The Status column in the Enrollment sheet will update to **registered** within a few seconds.



Screenshot: Enrollment sheet row for master finger with Status = registered

#### **X Warning**

Do not enroll the master finger with Role set to **employee**. If you do, it will record attendance instead of launching the WiFi portal, and it will not be usable for WiFi reconfiguration.

## 3.3 Verifying the Enrollment

After enrollment, confirm the following in the Enrollment sheet:

- The **FingerID** column now contains a number (e.g. 1).
- The **Status** column reads **registered**.
- The **Role** column reads **master**.

To confirm it works correctly, place the master finger on the sensor. The LED should flash yellow five times and then the device LED will turn purple — this means the WiFi portal has launched. Since you do not need to reconfigure WiFi right now, simply unplug and replug the device to restart it normally. The blue LED will return.



**✓ Tip**

Test the master finger immediately after enrollment so you know it works before you ever need it in an emergency.

## CHAPTER 4

## Enrolling Employees

Registering fingerprints for your team

### 4.1 How Enrollment Works

All enrollment is managed from the Enrollment tab in your branch's Google Sheet. You add a row with the employee's details and a **register** command, and the device detects it automatically within about 10 seconds. It then guides the employee through two finger placements, stores the fingerprint, and updates the sheet with the result.

#### ✓ Tip

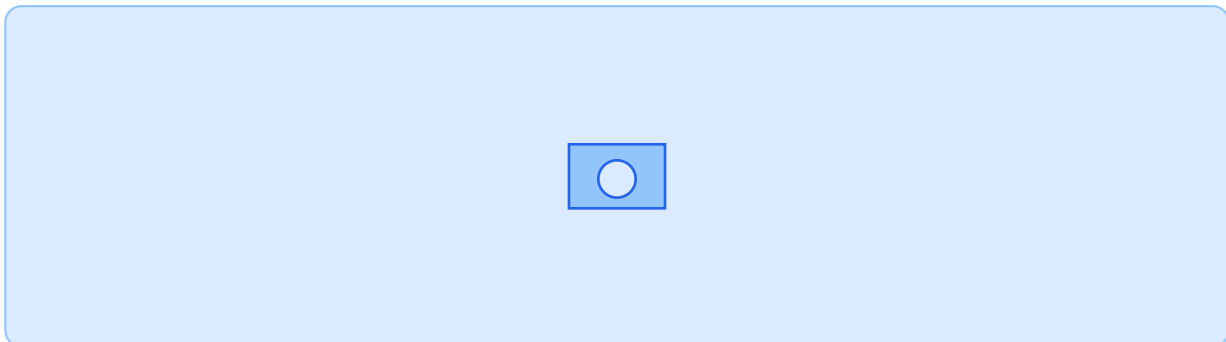
Have the employee standing next to the device before you add the row to the sheet, so they are ready when the yellow LED appears.

### 4.2 Registering a New Employee

- 1 Open the branch spreadsheet in Google Sheets and click the **Enrollment** tab.
- 2 Click on the first empty row below the last entry.
- 3 Fill in the columns as follows:

| Field     | Description                                                                  |
|-----------|------------------------------------------------------------------------------|
| FingerID  | Leave blank. The device fills this in automatically.                         |
| UniqueID  | The employee's Staff ID (e.g. <b>EMP-042</b> ). Must be unique per employee. |
| Name      | The employee's full name (e.g. <b>Kwame Mensah</b> ).                        |
| Role      | Type exactly: <b>employee</b>                                                |
| Command   | Type exactly: <b>register</b>                                                |
| Status    | Leave blank.                                                                 |
| Note      | Leave blank.                                                                 |
| CreatedAt | Optional — you may enter today's date for your records.                      |

- 4 Within about 10 seconds, the device LED will turn **solid yellow**, indicating it has picked up the enrollment job and is ready for the first scan.
- 5 Ask the employee to place their finger firmly and flatly on the sensor and hold it still.
- 6 The LED will go off briefly. Ask the employee to **lift their finger** and wait for the LED to turn yellow again.
- 7 Ask the employee to place the **same finger** on the sensor again for the second scan.
- 8 If successful, the LED flashes **green** and the Status column in the sheet updates to **registered** within a few seconds.



Screenshot: Enrollment sheet row after successful registration (Status = registered)

### 4.3 What Happens on the Device During Enrollment

| LED state                  | Meaning                                                                             |
|----------------------------|-------------------------------------------------------------------------------------|
| Solid yellow (first time)  | Waiting for the first finger placement.                                             |
| Solid yellow (second time) | First scan done — waiting for the same finger again.                                |
| Green flash                | Enrollment successful.                                                              |
| Red flash                  | Enrollment failed. The Status column will show the error. See Section 7.2 to retry. |

#### **X Warning**

Each employee must use the same finger every time they scan — both for enrollment and for all future attendance scans. Choose a finger with a clear, undamaged print. Index fingers usually work best.

### 4.4 Verifying a Successful Enrollment

Confirm the following before the employee walks away:

- The **FingerID** column contains a number.

- The **Status** column reads **registered**.

Ask the employee to do a test scan immediately. Their Time-In should appear as a new row in the Attendance sheet within a few seconds. If no row appears, see Chapter 7.

## CHAPTER 5

## Day-to-Day Operations

How the system runs once set up

### 5.1 How Employees Use the Device

Employee use is simple: place their registered finger on the sensor when arriving at the office, and place the same finger on the sensor when leaving. Nothing else is required.

- A **green flash** means the scan was accepted.
- A **red flash** means the fingerprint was not recognised. The employee should try again.
- The **first scan** of the day is recorded as **Time-In**.
- Every scan after that updates **Time-Out**. The last scan of the day is the final departure time.
- Once a scan is accepted, the employee must wait at least **60 seconds** before scanning again.

### 5.2 Monitoring Attendance in Real Time

Open the branch spreadsheet in Google Sheets at any time to see live attendance. New rows appear in the **Attendance** sheet within a few seconds of each scan. You do not need to refresh the page.

✓ **Tip**

Bookmark the branch spreadsheet URL for quick access during the work day.

### 5.3 Reading the Attendance Record

Each row in the Attendance sheet represents one employee's record for one day:

| Field         | Description                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------|
| Date          | The date of the record, in YYYY-MM-DD format.                                                   |
| Employee Name | The employee's full name.                                                                       |
| Staff ID      | The employee's unique staff ID.                                                                 |
| Time-In       | Time of the first scan of the day (HH:MM format).                                               |
| Time-Out      | Time of the last scan of the day (HH:MM format). Blank if the employee has not yet scanned out. |
| Status        | Computed automatically. See Section 5.4 below.                                                  |

**Note**

Column G is used internally by the system to prevent duplicate records. Do not delete, edit, or hide it.

## 5.4 Understanding Status Values

The Status column is computed automatically every time an employee scans, and is finalised by the nightly job at 11:30 PM.

| Status              | Meaning                                                              |
|---------------------|----------------------------------------------------------------------|
| On-Time             | Employee arrived and left within the allowed grace window.           |
| Late                | Employee's Time-In was after the allowed start grace period.         |
| Left-Early          | Employee's Time-Out was before the allowed end grace period.         |
| Late   Left-Early   | Employee arrived late and also left early.                           |
| No Check-Out        | Employee scanned in but did not scan out before the nightly job ran. |
| Late   No Check-Out | Employee arrived late and also did not scan out.                     |
| Absent              | No scan was recorded for the entire day.                             |

## CHAPTER 6

## Ongoing Admin Tasks

Everything you may need to do after first setup

### 6.1 Adding a New Employee

Follow the same process as Chapter 4. In brief:

- 1 Open the branch spreadsheet and go to the **Enrollment** tab.
- 2 Add a new row with the employee's Staff ID, Name, Role (**employee**), and Command (**register**).
- 3 Have the employee stand at the device and wait for the yellow LED.
- 4 Guide them through the two finger placements.
- 5 Confirm the Status reads **registered** before the employee leaves.

### 6.2 Removing an Employee

When an employee leaves the organisation, remove their fingerprint from the device so it can no longer be used for attendance.

- 1 Open the Enrollment tab of the branch spreadsheet.
- 2 Find the employee's row and note their **FingerID** number.
- 3 In that row, change the **Command** cell to **delete** and clear the **Status** cell.
- 4 The device will pick up the command within 10 seconds and delete the fingerprint. The Status column will update to **deleted**.

**Note**

Deleting a fingerprint does not remove the employee's historical attendance records. All past records remain in the Attendance sheet.

**X Warning**

Once deleted, the employee cannot scan in again until re-enrolled. Only delete employees who have genuinely left.

## 6.3 Replacing a Lost or Changed Fingerprint

If an employee can no longer be recognised by the sensor — due to a cut, injury, or switching to a different finger — delete their current fingerprint and register a new one.

- 1 Follow Section 6.2 to delete the existing fingerprint. Wait for the Status to update to **deleted**.
- 2 In the same row, change the Command back to **register**, clear the Status cell, and clear the FingerID cell.
- 3 Have the employee stand at the device and follow the enrollment steps in Chapter 4 from Step 4 onwards.

## 6.4 Changing the Office WiFi Network

If your office WiFi name or password changes, the device will lose its connection and the LED will glow **steady purple**. You will need the master finger to reconfigure it. This is why enrolling the master finger in Chapter 3 is important.

- 1 Make sure the device is powered on.
- 2 Place the **master finger** on the sensor. The LED will flash yellow five times, then turn purple — this means the WiFi setup portal is now running.
- 3 On your phone or laptop, connect to the **Attendance-Setup** WiFi network (password: **setup1234**).
- 4 Open a browser and go to **http://192.168.4.1**.
- 5 Enter the new WiFi name and password, then click **Save & Connect**.
- 6 The device will restart and the LED should return to **steady blue**.

### **X Warning**

If the master finger has not been enrolled yet and you need to change the WiFi, you will not be able to do so without contacting your system administrator. This is why you should always enroll the master finger first, as described in Chapter 3.

## 6.5 Reviewing the Monthly Email Report

On the first day of each month, an email is automatically sent to all configured admin addresses with the previous month's attendance summary. The email contains a section for each active branch, listing:

- Employees with absences, and how many days they were absent.



- Employees with late arrivals, and how many times.
- Employees with early departures.
- Employees with missing check-outs.

The email also includes an Excel (.xlsx) attachment for each branch. The attachment contains the full raw attendance log and a **Monthly Summary** tab with one row per employee showing all counts aggregated — useful for payroll or HR reviews.

✓ **Tip**

If you do not receive the monthly email, check your spam or junk folder first. If it is not there, contact your system administrator.

## 6.6 Correcting a Wrong Record

If a record needs to be corrected — for example, a Time-In value is wrong due to a clock error, or an employee's name was misspelled — you can edit the Attendance sheet directly.

- Click the cell you need to change and type the correct value.
- For Time-In and Time-Out, use the format **HH:MM** (e.g. **08:15**, **17:03**).
- For Date, use the format **YYYY-MM-DD** (e.g. **2026-04-08**).
- The system does not re-validate manually edited cells — whatever you type is saved as-is.

## 6.7 Checking for Missed Check-Outs

The nightly job at 11:30 PM automatically fills in **No Check-Out** for any employee who scanned in but never scanned out. If you want to check earlier in the day, open the Attendance sheet and look for rows where the **Time-Out** column is blank and the Date is today — those employees have arrived but not yet scanned out.

## CHAPTER 7

# Troubleshooting

Common problems and how to resolve them

## 7.1 Device LED Colour Guide

| LED state       | Meaning                                            | Action                                                        |
|-----------------|----------------------------------------------------|---------------------------------------------------------------|
| Steady blue     | Connected to WiFi, ready.                          | None — normal operation.                                      |
| Steady purple   | No WiFi connection.                                | Reconfigure WiFi using master finger (Section 6.4).           |
| Yellow flash x5 | Master finger detected — entering WiFi setup mode. | Proceed with setup if intended; otherwise restart the device. |
| Solid yellow    | Waiting for fingerprint during enrollment.         | Place the correct finger on the sensor.                       |
| Green flash     | Scan accepted / enrollment succeeded.              | None.                                                         |
| Red flash       | Finger not recognised, or enrollment failed.       | Try again. If persistent, re-enroll (Section 6.3).            |

## 7.2 Common Problems and Solutions

### ■ Scans are not appearing in the Attendance sheet

- Check the device LED is **steady blue**. If it is purple, the device has no WiFi — reconfigure using the master finger (Section 6.4).
- The device queues scans offline and sends them once WiFi is restored. After reconnecting, wait a minute and check if the queued scans appear.
- If the LED is blue but records still do not appear, contact your system administrator with the branch name and the time the scan was made.

### ■ An employee's finger is not being recognised

- Ask the employee to wipe the sensor surface with a clean dry cloth, then try again.
- Ensure the employee places the centre of their fingerprint on the sensor — not the tip of the finger.
- If the problem continues after several attempts, delete and re-enroll the fingerprint (Section 6.3).

### ■ Enrollment is stuck — the Status column never updates from 'processing'

- The device waits up to 20 seconds for a finger during enrollment. If no finger was placed in time, it times out. To retry: clear the Status cell and change the Command back to **register**.
- Confirm the device LED is **steady blue** (online) before retrying.
- Check the **Note** column — the device may have written an error code there.

### ■ The device is not picking up the enrollment command

- The device polls for new commands every 10 seconds. Wait at least 15 seconds after adding the row.
- Confirm the **Command** cell contains exactly **register**, **delete**, or **clearall** — no capital letters, no extra spaces.
- Confirm the **Status** cell is blank or contains **pending**. If it says **processing**, the device has already picked it up and you just need to wait.
- Confirm the device LED is **steady blue**.

### ■ The master finger is triggering the WiFi portal unexpectedly

- This is normal behaviour — placing the master finger on the sensor always launches the portal. To cancel, simply unplug and replug the device to restart it.
- If this keeps happening accidentally, consider enrolling the master finger using a less-used finger (e.g. little finger) to avoid accidental triggers.

### ■ The monthly summary email was not received

- Check your spam or junk folder.
- Confirm your email address was provided to your system administrator to be added to the recipient list.
- If neither helps, contact your system administrator.

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If a problem is not covered in this guide, contact your system administrator and provide: the branch name, a description of the issue, the LED colour or state at the time, and — if relevant — the exact text shown in the Status and Note columns of the Enrollment sheet.